uCNC RP2350B Motherboard

Generated board documents:

- Combined board review
- Front artwork and assembly layer
- Mirrored rear artwork
- Top 3D render
- Bottom 3D render

Board specification

Item	Specification
Product	LeanCam NCC / uCNC RP2350B Motherboard
Status	Placement prototype; copper routing intentionally deferred
Controller	Socketed WeAct Studio RP2350B Core Board
PCB outline	200 mm x 150 mm
PCB construction	Two copper layers, 1.6 mm finished thickness
Mounting	Four M3 clearance holes
Assembly	Main assembly on the front; J22 on the back
Input supply	24 V DC nominal machine supply, documented operating range 12-24 V
Main 5 V rail	LMR36510 synchronous buck, 1 A
Backlight rail	Isolated LMR36510 buck from PLC 24 V to `+12V_BL`
Logic supply	AMS1117-3.3, +3V3
Wi-Fi supply	Separate AMS1117-3.3, +3V3_WIFI
Analog supply	L78L12, +12V_AO
Isolated field supply	External +24 V through J33/J45
Board ID	MOD1, the socketed RP2350B module

The board is a machine-control carrier, not a safety-rated controller. Hazardous output power must be removed through an independent hard-wired safety circuit.

Digital I/O architecture

Function	Capacity	Implementation
Isolated PLC inputs	16	Four TLP291-4 quad optocouplers with four 4x10 k pull-up networks
Servo ZERO inputs	7	AX0-AX6, direct 3.3 V logic
Servo ERROR inputs	7	AX0-AX6, direct 3.3 V logic
Dedicated safety input	1	E-stop through a TLP291 SOP-4 optocoupler
General field outputs	16	DO0-DO15 through two TBD62083AFG packages
Motion outputs	7 axes	STEP/DIR per axis, shared `EN_3AX` and `EN_4AX` enables
Encoder inputs	3	A/B/Z channels with 5 V to 3.3 V conditioning
SD storage	1	Open-Smart 3.3 V TF breakout on J36
Analog inputs	3	0-10 V, resistor-divided into RP2350 ADC inputs

Analog outputs	4	0-10 V through TLV2374 and RC filtering

The digital chains are 32 bits wide:

- Input: four cascaded 74HC165 registers, eight bits each. U8/U9 carry PLC inputs 1-8 and 9-16. U6/U7 carry paired servo ZERO/ERROR signals.
- Output: four cascaded 74HC595 registers, eight bits each.
- Shift data uses GPIO20, GPIO21 and GPIO22; the input return is GPIO23.
- E-stop connects directly to MOD1 pin 53. HOLD and START were removed.

The complete GPIO and shift-bit mappings are in `documentation/gpio-map.md` and `documentation/io-bit-map.md`. Firmware must preserve the documented physical pin permutations.

Isolated field inputs

The 16 PLC inputs and dedicated E-stop form dry-contact loops powered from the external isolated 24 V field supply. J33 and J45 are electrically parallel two-pin access points for `+24V_FIELD` and the single `ISO_GND` return. The isolated field domain is not connected to motherboard GND.

Each PLC channel includes an indicator LED, 2.2 k series resistor, reverse-protection diode and one TLP291-4 optocoupler channel plus one element from a 4x10 k logic-side pull-up network. The E-stop pull-up remains discrete. Firmware must debounce the sampled contact.

Connectors

Ref	Function
J1	24 V DC power input
J2	Backlight/CCFL: `+12V_BL`, `ISO_GND`, protected 3.3 V PWM
J4	Physical 1x4 header; debug data pins unconnected
J5	Dedicated E-stop loop
J7, J8	16 isolated PLC input signals
J10, J11, J28, J29, J30, J46, J47	Seven 3-pin axis outputs: grouped EN, STEP, DIR
J12, J32	General field outputs DO0-DO15
J33	External isolated field +24 V supply and relay common return
J45	Second parallel +24 V field supply / return connector
J13-J15	Encoders 0-2
J40-J43, J48-J50	Seven paired ZERO/ERROR/GND input connectors
J18	5-pin hardware I2C1: SDA, SCL, 3.3 V, 5 V, GND
J36	1x9 header for Open-Smart 3.3 V TF Card Breakout
J20	Optional ESP-01/ESP-01S
J22	2x10, 2.00 mm LCD LVDS
J24	Three 0-10 V analog inputs
J25	Four 0-10 V analog outputs

Detailed connector pinouts are in `documentation/connectors.md`.

Power and protection

- Main input path: F1 2 A fuse, D1 24 V TVS and one 1 A SS16 series diode.
- D2 provides reverse-polarity protection for the logic-side input; D3 protects the isolated backlight-converter feed from J33/J45.

- U1 converts the protected machine input to the 1 A +5 V rail.
- U3 converts the isolated PLC 24 V field rail to `+12V_BL` at up to 1 A.
- The WeAct module receives +5 V and uses its own regulated 3.3 V rail.
- U2 independently supplies board logic, SD and LCD logic at +3V3.
- U34 independently supplies the optional ESP module at +3V3_WIFI.
- U37 supplies only the analog-output amplifier, avoiding a local hot spot.
- J33 and J45 provide parallel access to the external isolated +24 V field rail and its single `ISO_GND` return, used for dry-contact wetting and relay-output common.

Do not use the analog or isolated field rails to power motors, lamps or high-current solenoids. General field outputs are limited by their external supply and the two TBD62083AFG ratings.

Display and storage

- J22 carries three LVDS data lanes and one clock lane for a 2x10 panel interface. Rows 9-10 and pins 17-20 are not connected.
- RN26 and RN27 are fitted as the passive LVDS series path. The DS90LV047A driver U41 is removed.
- There is no HDMI connector or HDMI-specific support circuitry.
- The LCD connector has no backlight contact. J2 carries the isolated `+12V_BL`, `ISO_GND`, and a 3.3 V `BL_PWM` signal from GPIO35 through R131. The external backlight driver must provide PWM isolation.
- High-current backlight switching and the CCFL inverter remain external.
- The microSD card uses 3.3 V SPI and includes card-detect.

PCB and manufacturing data

Metric	Value
PCB footprints	274
Physical BOM components	270
Routed vias	112
Minimum configured track width	0.12 mm
Minimum drill	0.30 mm

The BOM is in `manufacturing/bom.csv`. This specification is informational and does not replace the KiCad schematic, PCB, DRC or ERC reports.

Verification status

The version documented here was checked with KiCad 10.0.1:

- DRC: 233 findings, including 124 open connections, 61 clearance findings, 56 silkscreen overlaps, 24 starved thermals and minor dangling copper.
- ERC: zero electrical-rule errors and 37 library-symbol cache warnings.
- The routed board is a working prototype variant, not a fabrication release.

These remaining findings must be reviewed before fabrication. The board is not declared production-ready or safety-certified.